

```
from google.colab import files
uploaded = files.upload()

import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread(list(uploaded.keys())[0])
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

laplacian_positive = np.array([
    [ 0, -1,  0],
    [-1,  4, -1],
    [ 0, -1,  0]
], dtype=np.float32)

laplacian_result = cv2.filter2D(gray, -1, laplacian_positive)

sharpened = cv2.add(gray, laplacian_result)

plt.figure(figsize=(14,5))

plt.subplot(1,3,1)
plt.title("Original Image")
plt.imshow(gray, cmap='gray')
plt.axis('off')

plt.subplot(1,3,2)
plt.title("Laplacian Output (Positive Center)")
plt.imshow(laplacian_result, cmap='gray')
plt.axis('off')

plt.subplot(1,3,3)
plt.title("Sharpened Image")
plt.imshow(sharpened, cmap='gray')
plt.axis('off')

plt.show()
```

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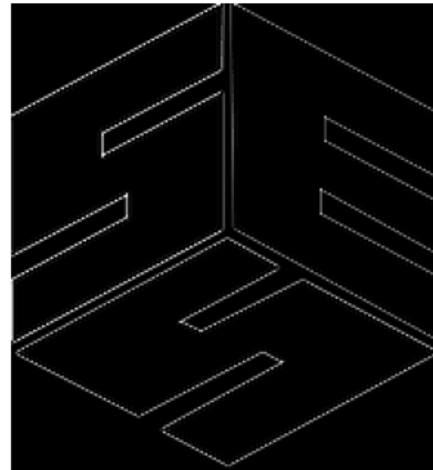
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Original Image



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Laplacian Output (Positive Center)



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Sharpened Image



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